

* We found that

$$\tau = \tau_0 \left(1 - \frac{11}{12} \alpha \left(\ln\left(\frac{s}{m^2}\right) + \mathcal{O}(m^0) \right) + \mathcal{O}(\alpha^2) \right)$$

where

$$\tau_0 = -g^2 (s^{-1} + t^{-1} + u^{-1}) \quad (\text{tree-level result.})$$

• Look at the massless limit, $m^2 \rightarrow 0$

So how do we deal with $\ln\left(\frac{s}{m^2}\right)$

A: We have made two mistakes!

i) We have assumed that we can isolate individual particles

→ Depended on the existence of an energy gap between one-particle states and the multiparticle continuum.

→ This gap vanishes if $\exists$ massless particles

• Here, this would imply the scattering process involved the creation of some extra low energy (soft) particles that escaped detection

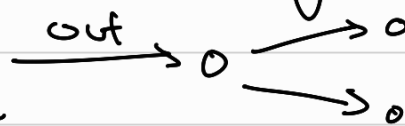
• Could have had extra soft particles in the initial scattering

• Final "particle" could have been multiple particles moving collinearly.

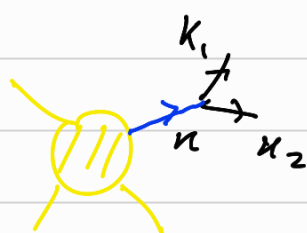
* τ : be amplitude for Scattering in φ^3 ($d=6$)

i) Consider

outgoing particle splits into two



$$k = k_1 + k_2$$



The amplitude for this process is

$$(ig) \left(\frac{-i}{k^2 + m^2} \right) \mathcal{T}$$

with k_1 and k_2 on-shell

- In the massless limit $\frac{1}{k^2 + m^2}$ can diverge

Cross-section of original theory:

$$|\mathcal{T}|^2 d\tilde{k}$$

- For the split we multiply $|\mathcal{T}_{\text{split}}|^2$ by $\frac{1}{2} d\tilde{k}_1 d\tilde{k}_2$ instead of $d\tilde{k}$
- $\frac{1}{2}$: Counts identical particles
- Assuming our detector cannot tell whether or not $\rightarrow \tilde{k}$, then we add the prob. of both events

$$|\mathcal{T}_{\text{obs}}|^2 d\tilde{k} = |\mathcal{T}|^2 d\tilde{k} + |\mathcal{T}_{\text{split}}|^2 \left(\frac{1}{2} d\tilde{k}_1 d\tilde{k}_2 \right) + \dots$$

- Can include a factor

$$1 = (2\pi)^{d-1} 2\omega \delta^{d-1}(k_1 + k_2 - k) d\tilde{k}$$

$$|\mathcal{T}_{\text{obs}}|^2 d\tilde{k} = |\mathcal{T}|^2 d\tilde{k} \left(1 + (2\pi)^{d-1} 2\omega \delta^{d-1}(k_1 + k_2 - k) g^2 \frac{1}{(k^2 + m^2)^2} \right)$$

and so

$$|\mathcal{T}_{\text{obs}}|^2 = |\mathcal{T}|^2 \left(1 + \frac{g^2}{(k^2 + m^2)^2} (2\pi)^{d-1} 2\omega \delta^{d-1}(k_1 + k_2 - k) \frac{1}{2} d\tilde{k}_1 d\tilde{k}_2 + \dots \right)$$

Now, in the massless limit

$$\begin{aligned} (k_1 + k_2)^2 &= k_1^2 + 2k_1 k_2 + k_2^2 \\ &= 2k_{01} k_{02} + 2\vec{k}_1 \cdot \vec{k}_2 \\ &= 2\omega_1 \omega_2 + 2\omega_1 \omega_2 \cos(\theta) \\ &= 2\omega_1 \omega_2 (\cos(\theta) - 1) \end{aligned}$$

$$C(\alpha + \beta) = C(\alpha)C(\beta) - \sin(\alpha)\sin(\beta)$$

$$C(2\theta) = C^2 - S^2 \\ = 1 - 2S^2$$

$$2S^2 = 1 - C(2\theta) \Rightarrow S^2(\theta) = \frac{1}{2}(1 - \cos(2\theta))$$

$$2\theta = \alpha \mid -2S^2(\theta/2) = \cos(\alpha) - 1$$

$\therefore$

$$K^2 = (k_1 + k_2)^2 = 2\omega_1 \omega_2 (-2S^2(\theta/2))$$

$$K^2 = -4\omega_1 \omega_2 S^2(\theta/2)$$

$$\text{For } m=0: \frac{1}{(K^2 + m^2)} \rightarrow \frac{1}{-4\omega_1 \omega_2 S^2(\theta/2)}$$

which can diverge for

$$\theta = \pi n, \quad n \in \mathbb{Z}$$

• Also for $m=0$

$$\frac{d^4 k_1}{(2\pi)^d 2k_1} \frac{d^4 k_2}{(2\pi)^d 2k_2} \sim \left(\frac{d^4 k_1}{(2\pi)^d 2k_1} \frac{d^4 k_2}{(2\pi)^d 2k_2} \right)$$

$$= \left(\frac{|\omega_1|^{d-1} d\omega_1}{(2\pi)^d 2\omega_1} \right) \left(\frac{|\omega_2|^{d-1} d\omega_2}{(2\pi)^d 2\omega_2} \right) \frac{d\Omega_{d-1}^{(1)}}{(2\pi)^{d-1}} \frac{d\Omega_{d-1}^{(2)}}{(2\pi)^{d-1}}$$



$$\sim \sin^{d-3}(\theta) d\theta \times (\quad)$$

$$\frac{d\tilde{k}_1 d\tilde{k}_2}{(K^2)^2} \sim \frac{d\omega_1}{\omega_1^{5-d}} \frac{d\omega_2}{\omega_2^{5-d}} \frac{d\theta}{\theta^{7-d}}$$

for small θ : $\sin(\theta) \approx \theta$

• Int. over each ω diverges at the low end

$$\left| \frac{d\omega}{\omega^{5-d}} \right| \sim [\omega]^{d-4}$$

$d \leq 4 \leftarrow$ diverges

• Int. over θ diverges for $d \leq 6$

For $d=6$ there is no problem with collinear particles (with small θ)

$$\theta < \delta$$

$$\text{Take the limit: } \frac{m^2}{K^2} \ll S^2 \ll 1$$

" " "

$$k_2 = k - k_1$$

integrate over $d^5 k_2$ $\beta = \tan(\frac{k_1}{k})$

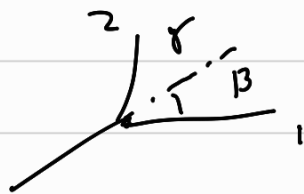
$$|k| = \frac{1}{2} \sqrt{s} \quad \text{in the limit } m \rightarrow 0$$

$$(2\pi)^5 2\omega \delta^5(k_1 + k_2 - k) \stackrel{w}{=} \tilde{d}k_1 \tilde{d}k_2$$

$$\rightarrow (\Omega_4 / 4(2\pi)^5) \frac{w}{\omega_1 \omega_2} |k_1|^4 d(k_1) \sin^3(\beta) d\beta$$

$\Omega_4 = 2\pi^2$ (area of unit 4-sphere S^4)

• Let γ be the angle between k_2 and k



$$\theta = \gamma + \beta$$

$$\sin(\theta) |k_1| = \sin(\gamma) |k|$$

$$\sin(\theta) |k_2| = \sin(\beta) |k|$$

$$\beta = x\theta \quad \text{and} \quad \gamma = (1-x)\theta$$

$$0 \leq x \leq 1 \quad \text{and} \quad \theta \leq \delta < \pi$$

$$\bullet \quad k^2 + m^2 \approx -x(1-x) \vec{k}^2 \left[\theta^2 + \left(\frac{m^2}{\vec{k}^2} \right) f(x) \right]$$

$$\text{where } f(x) = (1-x+x^2) / (x-x^2)^2$$

$$\omega_1 = |k_1| = (1-x)|k|$$

$$\omega_2 = |k_2| = x|k|$$

• Change m.t. variables from $(|k|, \beta) \rightarrow (x, \theta)$

$$|\tau_{\text{abs}}|^2 = |\tau|^2 \left[1 + \frac{g^2 \Omega_4}{4(2\pi)^5} \int_0^1 x(1-x) \int_0^\delta \frac{\theta^3 d\theta}{\left[\theta^2 + \frac{m^2}{\vec{k}^2} f(x) \right]^2} + \dots \right]$$

$$= |\tau|^2 \left(1 + \frac{1}{12} \alpha \left(\ln \left(\delta^2 \vec{k}^2 / m^2 \right) + c \right) + \dots \right)$$

$$c = (4 - 3\sqrt{3}\pi) / 3 = -4.11$$

• Do the same for all other (three) external particles: $2/12 \rightarrow 4/12$

$$|\tau_{\text{abs}}|^2 = |\tau|^2 \left(1 + \frac{4}{12} \alpha \left(\ln \left(\frac{\delta^2 \vec{k}^2}{m^2} \right) + c \right) + \mathcal{O}(\alpha^2) \right)$$

$$\vec{k}^2 = \frac{1}{4} s$$

Now, since we have $\tau = \tau_0 \left(1 - \frac{11}{12} \alpha \left(\ln \left(\frac{s}{m^2} \right) + \mathcal{O}(m_0) \right) \right)$

$$|\tau_{\text{abs}}|^2 = |\tau_0|^2 \left(1 - \frac{11}{12} \alpha \left(\ln \left(\frac{s}{m^2} \right) + \mathcal{O}(m_0) \right) \right)^2 \left(1 + \frac{4}{12} \alpha \left(\ln \left(\frac{\delta^2 |\vec{k}_s|}{m^2} \right) + c \right) \right)$$

$$= |\tau_0|^2 \left[1 - \alpha \left(\frac{3}{2} \ln(s/m^2) + \frac{1}{3} \ln(\delta^2/s^2) + \mathcal{O}(m_0) \right) + \mathcal{O}(\alpha^2) \right]$$

$\ln(\frac{1}{\delta^2})$: depends on the properties of our detector
 $m^2/k^2 \ll \delta^2 \ll 1$

$\ln(s/m^2)$: already seen and blows up at $m \rightarrow 0$

→ This means there is some other mistake in our analysis